Navigating through the **Azure Messaging** (over)choice

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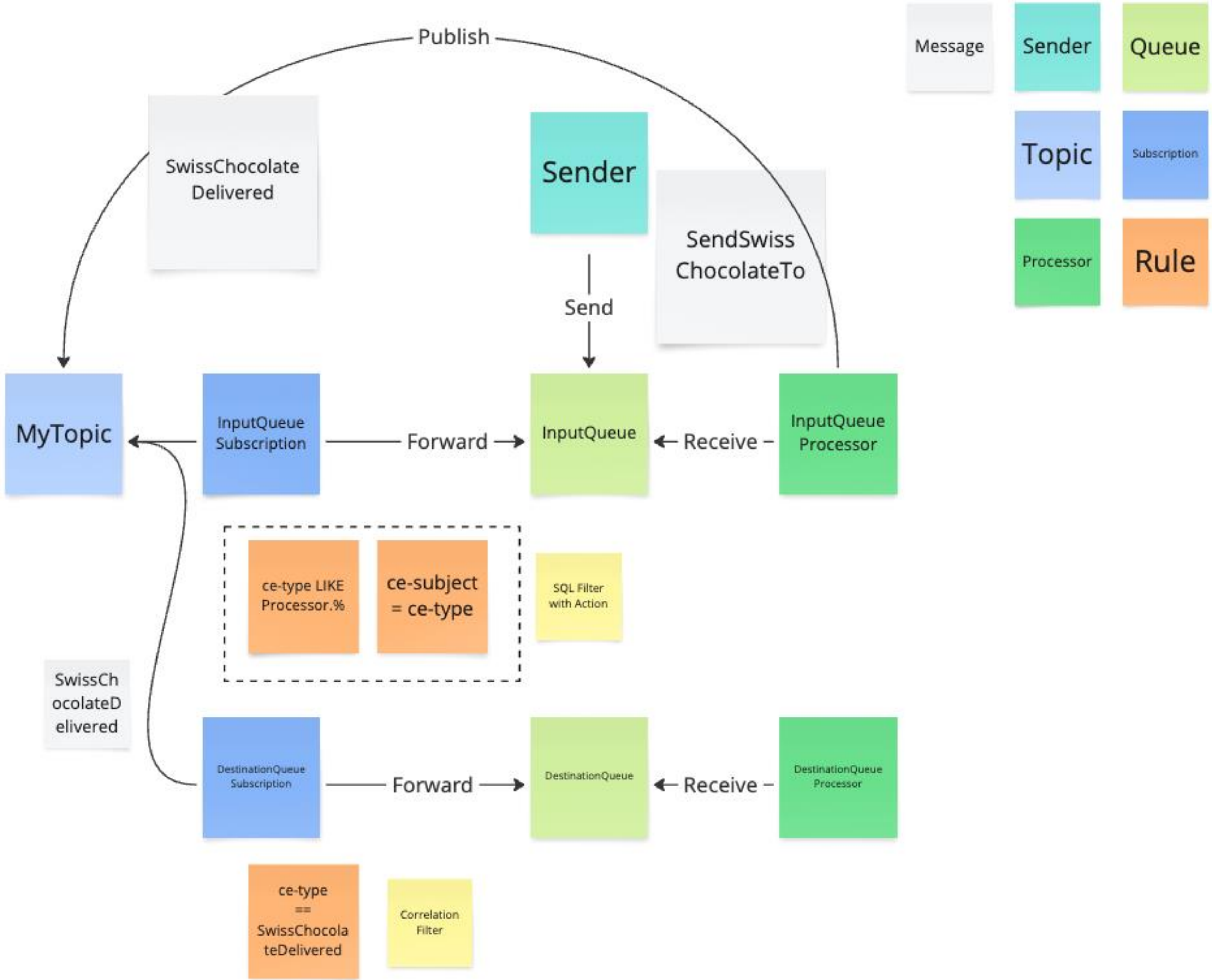
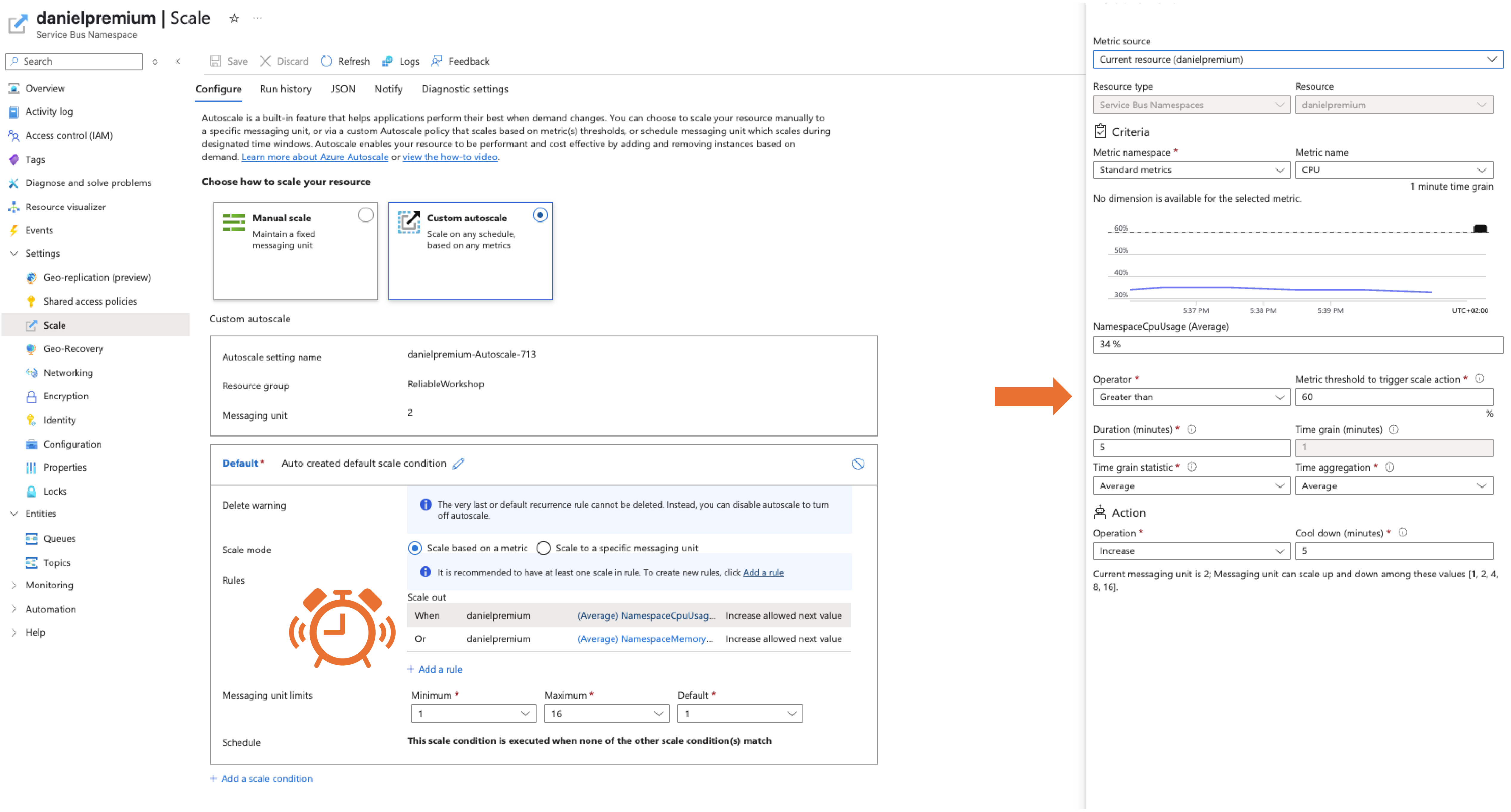
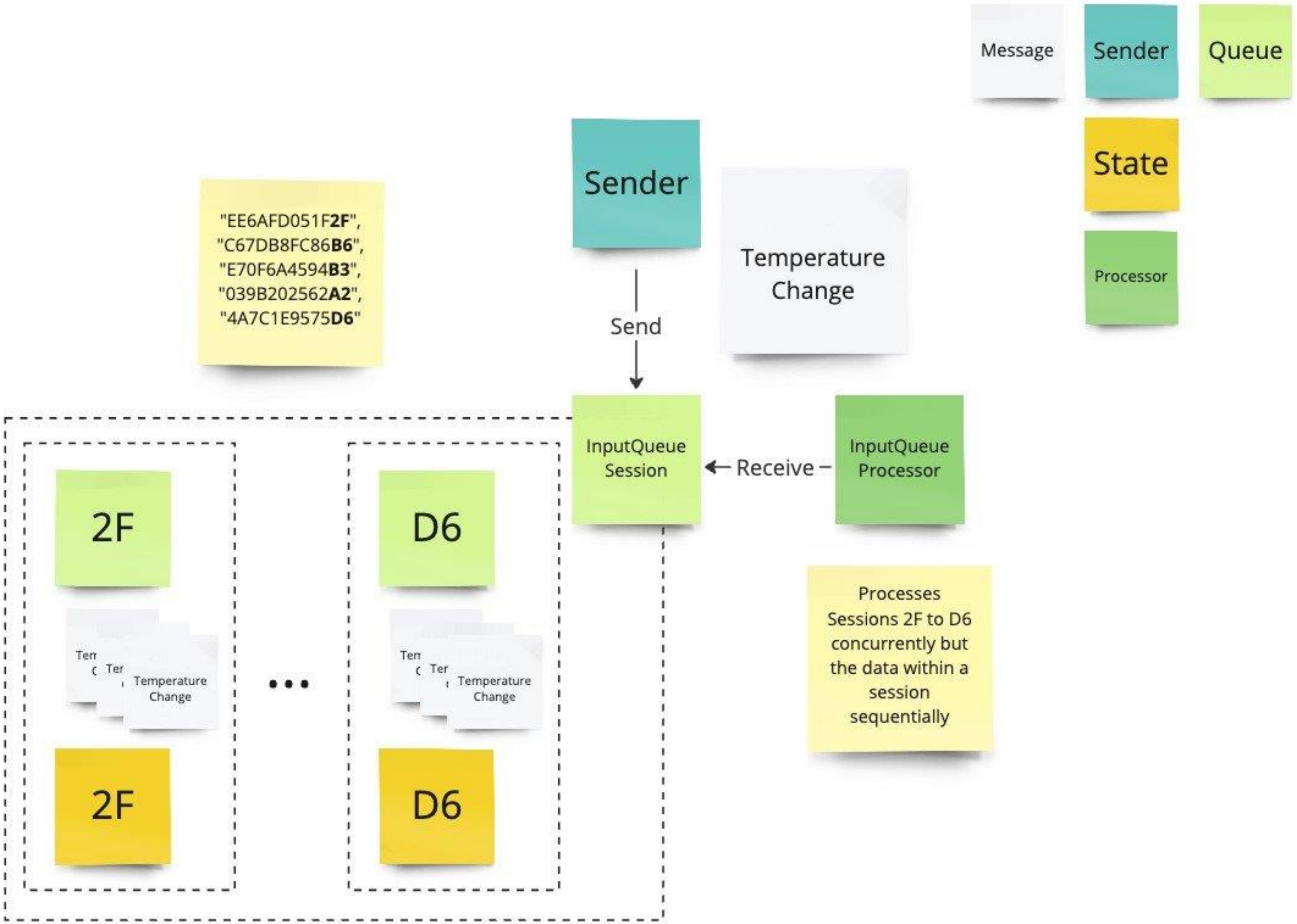

Message
Sender
Queue
Topic
Subscription
Processor
Rule
Publish
SwissChocolate Delivered
Sender
Send
SendSwiss ChocolateTo
MyTopic
InputQueue Subscription
Forward
InputQueue
Receive
InputQueue Processor
ce-type LIKE Processor.%
ce-subject = ce-type
SQL Filter with Action
SwissCh ocolateD elivered
DestinationQueue Subscription
Forward
DestinationQueue
Receive
DestinationQueue Processor
ce-type == SwissChocola teDelivered
Correlation Filter

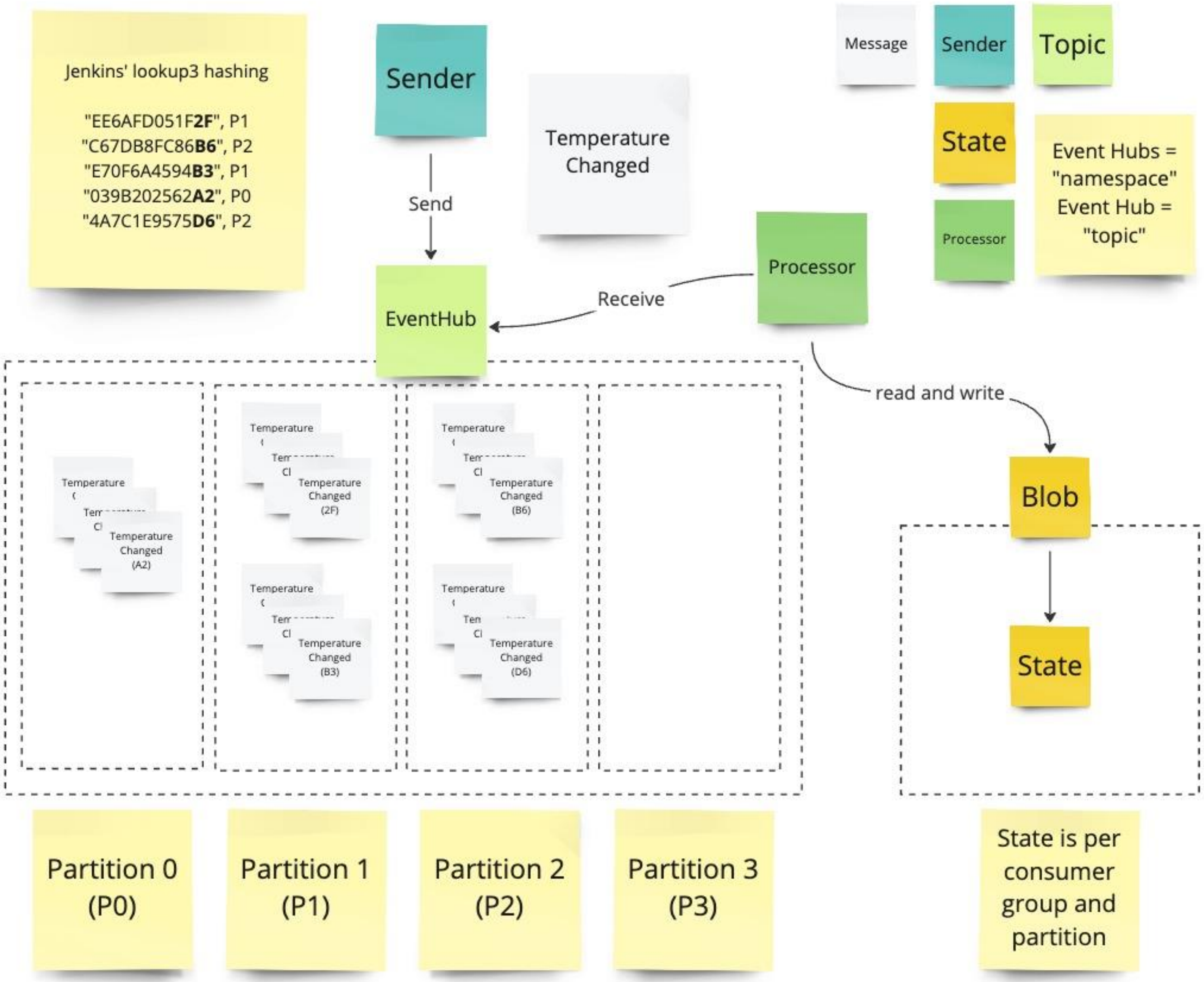
CODE & DEMOS


danielpremium | Scale
Service Bus Namespace
Search
Save
Discard
Refresh
Logs
Feedback
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Events
Settings
Geo-replication (preview)
Shared access policies
Scale
Geo-Recovery
Networking
Encryption
Identity
Configuration
Properties
Locks
Entities
Queues
Topics
Monitoring
Automation
Help
Configure
Run history
JSON
Notify
Diagnostic settings
Autoscale is a built-in feature that helps applications perform their best when demand changes. You can choose to scale your resource manually to a specific messaging unit, or via a custom Autoscale policy that scales based on metric(s) thresholds, or schedule messaging unit which scales during designated time windows. Autoscale enables your resource to be performant and cost effective by adding and removing instances based on demand. Learn more about Azure Autoscale or view the how-to video.
Choose how to scale your resource
Manual scale
Maintain a fixed messaging unit
Custom autoscale
Scale on any schedule, based on any metrics
Custom autoscale
Autoscale setting name
danielpremium-Autoscale-713
Resource group
ReliableWorkshop
Messaging unit
2
Default *
Auto created default scale condition
Delete warning
The very last or default recurrence rule cannot be deleted. Instead, you can disable autoscale to turn off autoscale.
Scale mode
Scale based on a metric
Scale to a specific messaging unit
It is recommended to have at least one scale in rule. To create new rules, click Add a rule
Rules
Scale out
When
danielpremium
(Average) NamespaceCpuUsag...
Increase allowed next value
Or
danielpremium
(Average) NamespaceMemory...
Increase allowed next value
+ Add a rule
Messaging unit limits
Minimum *
1
Maximum *
16
Default *
1
Schedule
This scale condition is executed when none of the other scale condition(s) match
+ Add a scale condition
Metric source
Current resource (danielpremium)
Resource type
Service Bus Namespaces
Resource
danielpremium
Criteria
Metric namespace *
Standard metrics
Metric name
CPU
1 minute time grain
No dimension is available for the selected metric.
60%
50%
40%
30%
5:37 PM
5:38 PM
5:39 PM
UTC+02:00
NamespaceCpuUsage (Average)
34 %
Operator *
Greater than
Metric threshold to trigger scale action *
60
%
Duration (minutes) *
5
Time grain (minutes)
1
Time grain statistic *
Average
Time aggregation *
Average
Action
Operation *
Increase
Cool down (minutes) *
5
Current messaging unit is 2; Messaging unit can scale up and down among these values [1, 2, 4, 8, 16].

Performance
Add metric
Add filter
Apply splitting
Line chart
Drill into Logs
New alert rule
Save to dashboard
danielpremium, CPU, Avg
danielpremium, Memory Usage, Avg
danielpremium, Throttled Requests., Sum
danie
Service Bu
Search
Overview
Activity log
Access contr
Tags
Diagnose an
Resource visi
Events
Settings
Geo-repl
Shared a
Scale
Geo-Rec
Network
Encryptic
Identity
Configur
Propertie
Locks
80%
70%
60%
50%
40%
30%
20%
10%
0%
Finally!
redeploy
redeploy
redeploy
4:30
4:45
UTC+02:00
UTC+02:00
2025 17:02:03 G...
CPU (Avg), danielpremium 30.1500%
Memory Usage (Avg), danielpremium 10.0167%
Throttled Requests. (Sum), danielpremium 0

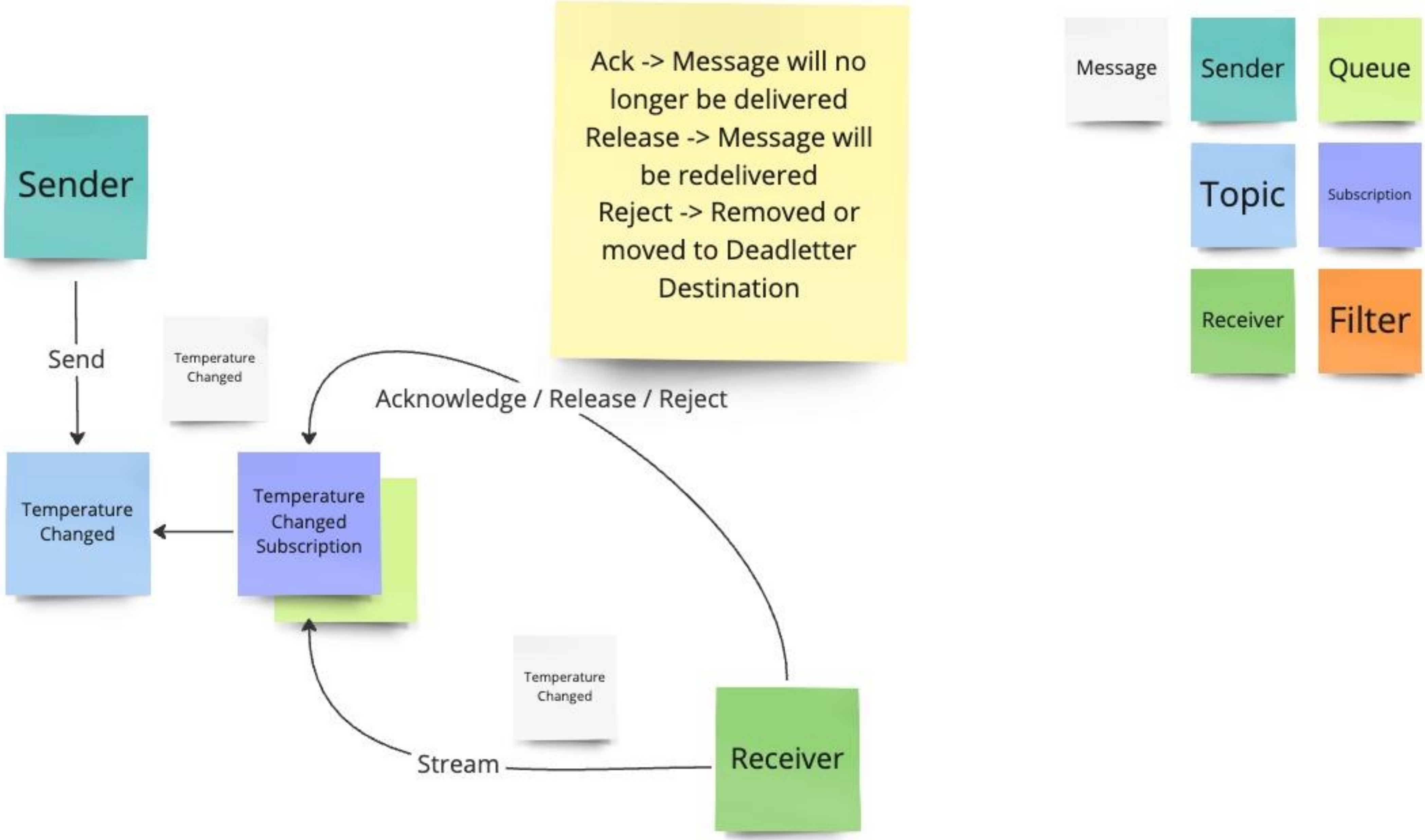

Message
Sender
Queue
State
Processor
"EE6AFD051F2F",
"C67DB8FC86B6",
"E70F6A4594B3",
"039B202562A2",
"4A7C1E9575D6"
Sender
Temperature Change
Send
InputQueue Session
Receive
InputQueue Processor
2F
Temperature Change
2F
D6
Temperature Change
D6
Processes Sessions 2F to D6 concurrently but the data within a session sequentially

CODE & DEMOS

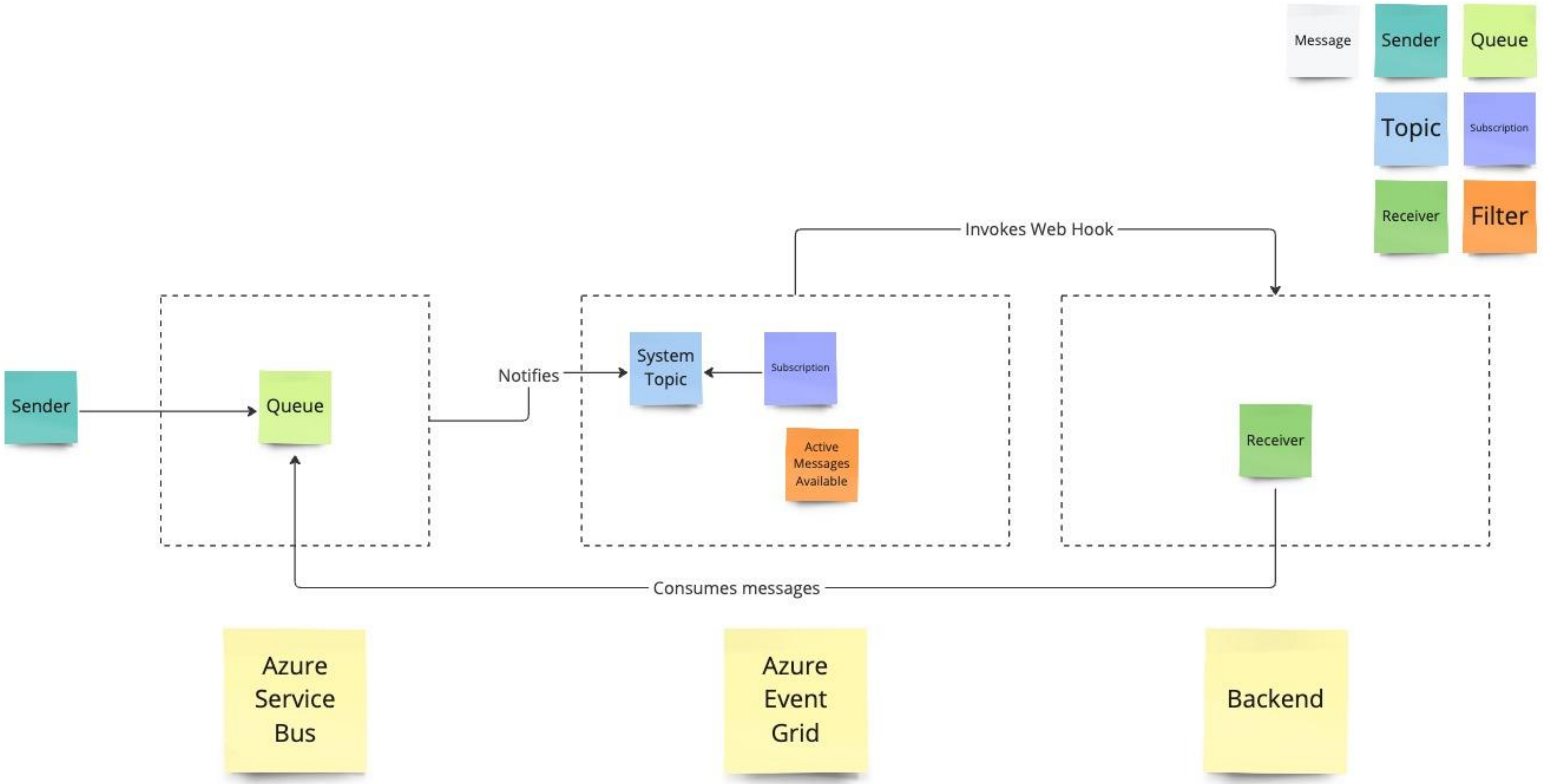


Jenkins' lookup3 hashing
"EE6AFD051F2F", P1
"C67DB8FC86B6", P2
"E70F6A4594B3", P1
"039B202562A2", P0
"4A7C1E9575D6", P2
Sender
Send
Temperature Changed
EventHub
Receive
Processor
read and write
Message
Sender
Topic
State
Processor
Event Hubs = "namespace" Event Hub = "topic"
Temperature Changed (A2)
Temperature Changed (2F)
Temperature Changed (B3)
Temperature Changed (B6)
Temperature Changed (D6)
Blob
State
Partition 0 (P0)
Partition 1 (P1)
Partition 2 (P2)
Partition 3 (P3)
State is per consumer group and partition

CODE & DEMOS


Sender
Send
Temperature Changed
Temperature Changed
Temperature Changed Subscription
Acknowledge / Release / Reject
Ack -> Message will no longer be delivered
Release -> Message will be redelivered
Reject -> Removed or moved to Deadletter Destination
Temperature Changed
Stream
Receiver
Message
Sender
Queue
Topic
Subscription
Receiver
Filter


CODE & DEMOS



Message
Sender
Queue
Topic
Subscription
Receiver
Filter
Invokes Web Hook
Sender
Queue
Notifies
System Topic
Subscription
Active Messages Available
Receiver
Consumes messages
Azure Service Bus
Azure Event Grid
Backend

CODE & DEMOS

Segment	Eventing PubSub	Streaming	Enterprise Messaging
Product	Event Grid	Event Hubs	Service Bus
What do you care about	• Communication between apps / orgs • Individual message • Push and pull semantics • Filtering and routing • Pay as you go • Fan out	• Many messages in a Stream (think in MBs) • Ease of use and operation • Low cost • Fan in • Strict ordering • Kafka compliant • Works with other tools	• Instantaneous consistency • Strict ordering • JMS • Non-repudiation & Security • Geo-Replication & Availability • Rich features (de-dupe, scheduling, etc.)
What you care less about	• Ordering of messaging • Instantaneous consistency	• Individual message semantics • Server-side cursor • Only Once	• Cost • Simplicity
Focus point	• Reliable fan out push at massive scale	• High scale distributed log	• Highly indexed full featured message broker
	Serverless	Streaming Data	Enterprise

Use all of them

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